Predicting High School Dropout Risk Using HSLS:09

Capstone Project Proposal • Data Science Program

March 2026

1. Project Title

Early Warning for High School Dropout: A Machine Learning Approach Using the HSLS:09 Longitudinal Study

2. Problem Statement

High school dropout carries lasting consequences for students, including reduced earnings and limited access to further education. The primary stakeholders for this work are school counselors, administrators, and policymakers who need actionable, data-driven tools to identify at-risk students early and allocate support effectively. This project focuses on whether we can use data to predict dropout risk early enough for schools to intervene effectively.

3. Dataset

The dataset used in this project is the

High School Longitudinal Study of 2009 (HSLS:09), produced by the National Center for Education Statistics (NCES) within the U.S. Department of Education's Institute of Education Sciences (IES). Each row in the dataset represents one individual student in the HSLS study.

Key characteristics of the dataset include:

- A nationally representative cohort of 9th-grade students first surveyed in fall 2009, with follow-up surveys in 2012 (First Follow-up), 2013 (Update), and 2016 (Second Follow-up).
- More than 25,000 students across roughly 940 participating high schools in 50 states and the District of Columbia
- Multi-source data: student assessments and questionnaires, parent surveys, high school transcripts, and postsecondary transcript and financial aid records. While the full HSLS:09 dataset also includes teacher, counselor, and administrator questionnaires, this project focuses exclusively on the student-level file.
- More than 3,000 variables covering demographics, academic performance, motivation, family background, school context, and postsecondary outcomes.
- Structured reserve codes distinguishing not-applicable responses from true missing data, allowing for principled preprocessing.

The public-use file is accessible via the NCES Online Codebook and DataLab. The target variable for this project is **`X4EVERDROP`**, a binary composite indicator (0 = No, 1 = Yes) that records whether the student ever had a known dropout episode by the second follow-up in 2016.

4. Objectives of the Analysis

This project aims to answer the following research questions and generate the following insights:

1. **Primary prediction objective:** Can dropout risk be predicted with meaningful accuracy using only base-year (9th grade) data, and which feature domains contribute most to predictive performance?
2. **Feature horizon comparison:** Does including first follow-up (11th grade) information improve predictive performance compared with using only base-year data?
3. **Feature importance:** Which categories of student-level factors - such as academic performance, engagement and behavior, school attitudes and motivation, family and socioeconomic background, school support and guidance, among others - are the strongest predictors of dropout risk?

5. Planned Methodology

This project will follow a supervised learning workflow to predict student dropout risk using the HSLS dataset.

- First, the data will be loaded, inspected, and cleaned. Restricted public-use variables suppressed with `-5` will be removed, and other reserve codes such as `-6`, `-7`, `-8`, and `-9` will be treated as missing values and handled appropriately. A data dictionary will also be created to document variable meanings and cleaning decisions.
- Next, exploratory data analysis (EDA) will be conducted to understand the target variable, examine class balance, identify missing data patterns, and explore relationships between dropout and key categories of variables such as academic performance, engagement and behavior, school attitudes, educational expectations, and family socioeconomic background and more.
- After EDA, relevant features will be selected and, where useful, new features will be created to improve prediction. A baseline supervised learning model will then be built using a train/test split. After establishing baseline performance, at least one improved model will be tested and compared fairly against the baseline.
- Model evaluation will focus on appropriate classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, with special attention to class imbalance and potential data leakage. Finally, the best-performing model will be selected, interpreted, and presented using clear visualizations and tables to communicate the most important predictors of dropout risk.

6. Expected Outcomes

By the end of this project, we expect to deliver the following results and insights:

- A trained, calibrated dropout risk model capable of generating a probability score between 0 and 1 for each student, representing their likelihood of dropping out.
- A clear identification of the most important factors associated with dropout risk, such as academic performance, engagement and behavior, educational expectations, and socioeconomic background.
- A comparison of prediction performance across two data horizons (9th grade only vs. full longitudinal) to quantify the value of additional data collection.

7. Division of Responsibilities

The following is a tentative breakdown of responsibilities. Roles may shift as the project evolves and as we better understand the depth of each component.

Team Member	Primary Roles	Secondary Role
Nafe Abubaker	Project coordinator, Data lead	Modeling lead
Afaf Amwas	Project coordinator, Data lead	EDA and visuals lead
Tala Hilqawi	Slides lead, EDA and visuals lead	Documentation lead
Ayham Al-Duwairi	Modeling lead, Documentation lead	GitHub lead
Heba Qasim	EDA and visuals lead, Documentation lead	Slides lead
All Members	Work together as a team	Whatever needed.

This proposal is tentative. All decisions — including the feature set, models, and team structure — may evolve as we explore the data and receive feedback